

Family Lymantriidae

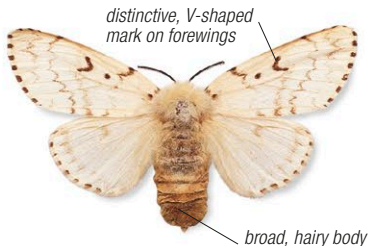
Tussock moths

Occurrence 2,600 spp. worldwide; on foliage of deciduous and coniferous trees, shrubs



Tussock moths resemble noctuids (see right) and although most are dull in color, tropical species can be colorful. The wingspan is $\frac{3}{4}$ –2 $\frac{1}{4}$ in (2–6 cm); males are slightly smaller than the females, which are sometimes wingless. Females lay eggs in clumps on the bark of host trees and shrubs. Adults lack a proboscis and do not

feed. The caterpillars are hairy, often brightly colored, and feed in groups on foliage. They may induce an allergic reaction if touched.



LYMANTRIA DISPAR
Commonly called the gypsy moth, this species is a native of Europe and Asia. Introduced to North America to produce inexpensive silk, it escaped from captivity and has become a pest.

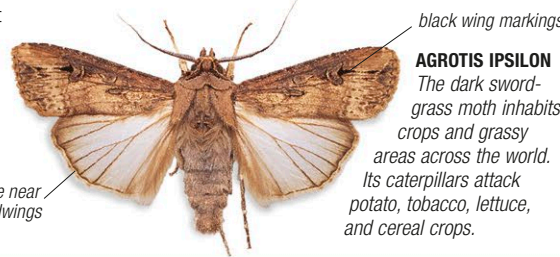
Family Noctuidae

Noctuid moths

Occurrence 35,000 spp. worldwide; on all types of vegetation, including crops



Almost every important crop plant in the world is attacked by one or more noctuid moth species. These moths have narrow forewings



AGROTIS IPSILON
The dark sword-grass moth inhabits crops and grassy areas across the world. Its caterpillars attack potato, tobacco, lettuce, and cereal crops.

Family Papilionidae

Swallowtails

Occurrence 600 spp. worldwide in warmer regions; in flower-rich, open or shaded areas

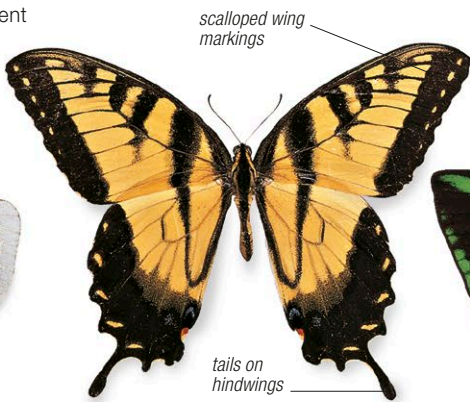


This family includes birdwings—the world's largest butterflies, now protected by law. True to their name, many swallowtail species have tails on their hindwings. The wings are typically dark with bands, spots, or patches of white, yellow, orange, red, green, or blue, and wingspan ranges from 1 $\frac{3}{4}$ –11 in (4.5 to 28 cm). Round eggs are laid singly on host plants and pupation also occurs here. The

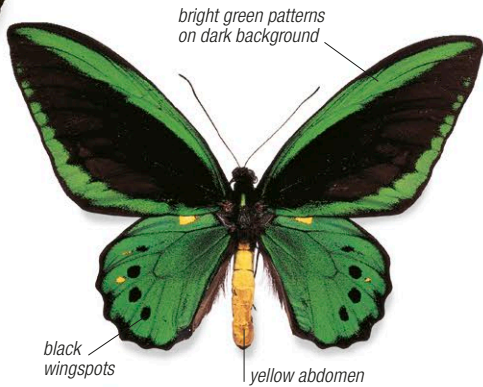
caterpillars have a forked, fleshy scent gland, which emits an odor that protects them from predators. They are herbivorous and eat a range of foliage. Adult swallowtails feed on nectar and other liquids.



PARNASSIUS APOLLO
Also known as the Apollo, this species is found in mountainous areas of Europe and parts of Central Asia.



PAPILIO GLAUCUS
Commonly called the tiger swallowtail, this North American species inhabits woodland and gardens. Its caterpillars feed among high foliage in many deciduous trees.



ORNITHOPTERA PRIAMUS
Also called the Cairns birdwing, this species inhabits tropical forests in the Moluccas, Papua New Guinea, the Solomon Islands, and northeastern Australia. Females are larger than males, and are drably colored.

Family Nymphalidae

Nymphalid butterflies

Occurrence 6,000 spp. worldwide; in flower-rich meadows, woodland clearings

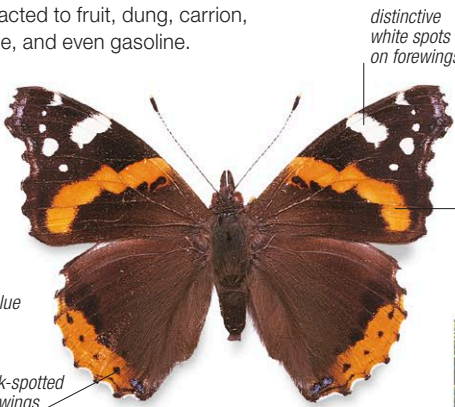


This family includes some of the world's most eye-catching butterflies, their most characteristic feature being their very short, brushlike front legs, which have a sensory function. These are held off the ground, close to the head. The wingspan

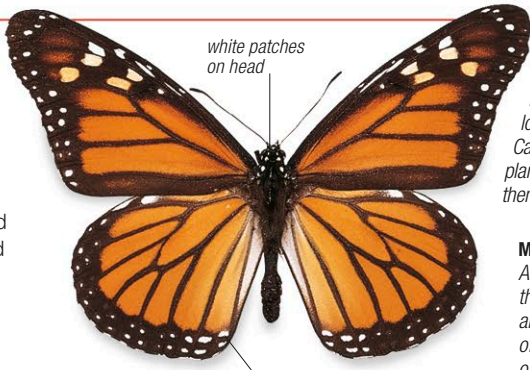
is 1 $\frac{1}{4}$ –6 in (3–15 cm); the upper wing surfaces are brightly colored but the undersides have cryptic coloring to camouflage the butterfly at rest. Males and females may be differently colored, and there may be seasonal variation. Females lay groups of rounded eggs on the foliage of trees, shrubs, and herbaceous plants. The pupae often have warty, conspicuous bumps and hang from the host plant by a group of terminal hooks called cremasters. When young, the caterpillars may feed communally on foliage. Adults feed on nectar and liquids; some are attracted to fruit, dung, carrion, urine, and even gasoline.



NYMPHALIS ANTIOPA
Called the Camberwell beauty in the UK, and the mourning cloak in North America, this butterfly inhabits grassland in the Northern Hemisphere.



VANESSA ATALANTA
The red admiral is widespread in the Northern Hemisphere. Its caterpillars feed on nettles and hops.



DANAUS PLEXIPPUS
The monarch butterfly is famous for its long-distance migrations, the longest of which extends from Mexico to Canada. Its caterpillars feed on milkweed plants; a substance in these plants makes them unpalatable to predators.



MORPHO PELEIDES
Also called the common blue morpho, this butterfly inhabits forests in South and Central America. With a wingspan of 3 $\frac{3}{4}$ –4 $\frac{3}{4}$ in (9.5–12 cm), it is one of the larger nymphalids. Its caterpillars can produce a noxious defensive odor.